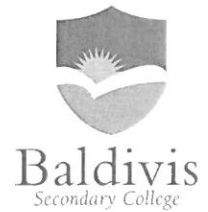


Mini Test 3- Year 9
Ratio and Proportion



Resources allowed: Calculator and A4 notes page one side

Name: _____

Class: _____

Time: 25 minutes

Total Marks: 23

Multiple Choice Section: Please circle the correct answer.

(7 marks)

1. What speed is required to run 100 m in 10 seconds?
A 5 m/s ☒ B 10 m/s C 20 m/s D 25 m/s

2. It takes 1 1/2 minutes to put 30 L of petrol in my car. At what rate is the petrol flowing?
☒ A 20 L/min B 30 L/min C 45 L/min D 90 L/min

3. Cherries are on sale for \$6.50 a kilogram. How many kilograms of cherries can I buy for \$10?
A 0.65 kg B 0.94 kg C 1.26 kg ☒ D 1.54 kg

4. Which of the following is the best buy?
A 500 g box for \$1.35 B 1 kg box for \$2.50 ☒ C 5 kg bag for \$11.25 ~~D 10 kg bag for \$23.00~~

5. Tony has 24 marbles and Joni has 18. What is the simplest way to express the ratio of Tony's to Joni's marbles?
A 24 : 18 B 12 : 9 C 8 : 6 ☒ D 4 : 3

6. Divide \$200 in the ratio 2 : 3 : 5.
A \$20, \$30 and \$50 B \$20, \$45 and \$135 ☒ C \$40, \$60 and \$100 D \$50, \$60 and \$90

7. 10 : 15 and which ratio below are inversely proportional?
A 2 : 3 B 4 : 3 C 3 : 4 ☒ D 9 : 6

Short Answer Calculation Section.

8. In a scale drawing the length and width of a block of land are 10 mm and 6 mm respectively. If the real width of the land is 15 m, how long is it? (3 marks)

$$15\,000 \div 6 = 2500$$

$$2500 \times 10 = 25\,000\text{mm}$$
$$= 25\text{m}$$

9. Sam can paint two houses in a day, whereas it takes Roger two days to paint one house. If they work together, how long will it take for them to paint a house? (3 marks)

$$S: 2 \text{ h/d} \quad R: \frac{1}{2} \text{ h/d} \quad \text{Together: } 2\frac{1}{2} \text{ h/d}$$

$$\text{days per house} = 24 \div 2\frac{1}{2} = 9.6 \text{ hrs}$$

$$\text{or} = 1 \div 2\frac{1}{2} = 0.4 \text{ days}$$

10. A bus has a flat fee to hire it. It will cost \$50 per person if there are 20 people on the charter bus. How many people would be needed to reduce the cost to \$40 per person? (2 marks)

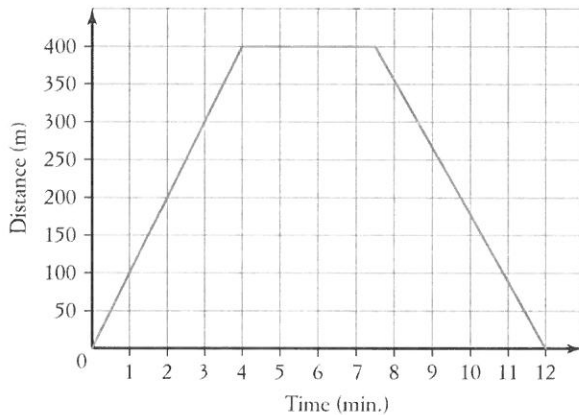
$$\text{Total cost} = 50 \times 20 = \$1000$$

$$\text{People} = \text{Cost} \div \text{cost/person}$$

$$= 1000 \div 40$$

$$= 25 \text{ people}$$

11. The travel graph below describes Salah's trip from home to the milk bar and back. (5 marks)



- a How far is the shop from home?

400m

- b How long did she take to get to the milk bar?

4 mins

- c What speed did she walk to the milk bar?

$$21 = 100 \text{ m/min}$$

$$400 \div 4 = 100$$

- d How long did she spend at the milk bar?

$$7\frac{1}{2} - 4 = 3\frac{1}{2} \text{ minutes}$$

- e What speed did she walk home from the milk bar?

$$12 \div 7\frac{1}{2} = 4\frac{1}{2} \text{ mins} \quad 400 \div 4.5 = 89 \text{ m/min}$$

12. An old puzzle goes as follows. If a hen and a half can lay an egg and a half in a day and a half, how long will it take a hen to lay a dozen eggs? You may assume half an egg means half the time to produce an egg. (3 marks)

$$1.5 \text{ hens} = 1.5 \text{ eggs (per 1.5 days)}$$

$$1 \text{ hen} = 1 \text{ egg (per 1.5 days)}$$

$$1 \text{ hen} = \frac{2}{3} \text{ egg (per day)}$$

$$12 \text{ eggs} \div \frac{2}{3} \text{ egg/day} = \text{days}$$

$$= 18 \text{ days}$$

End Test